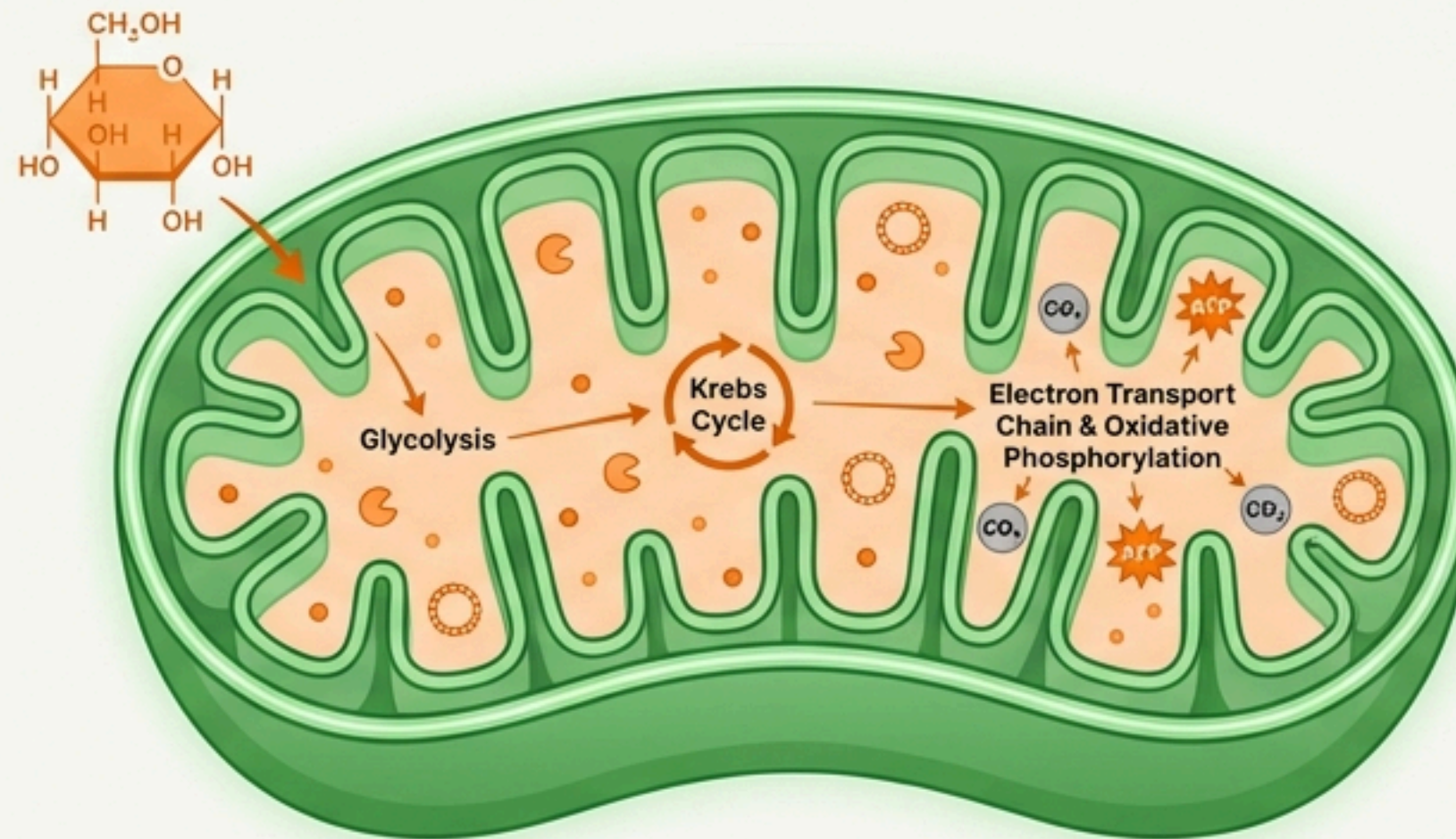


Respiration in Plants

(Class 11, Chapter 12)



100%
NCERT
Content

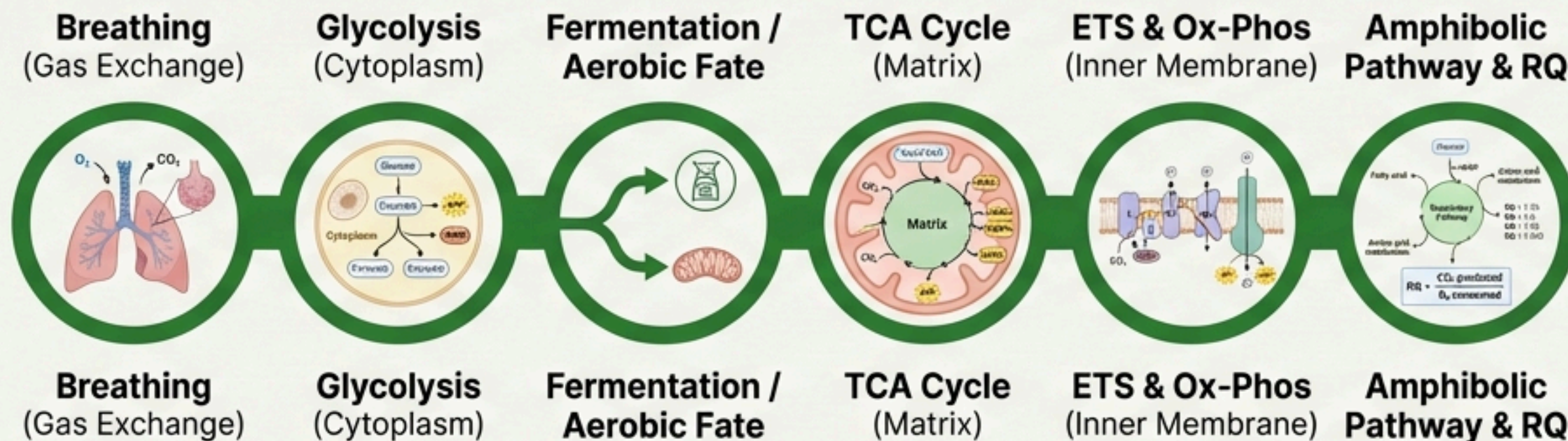
Complete NCERT Analysis & High-Yield
Revision for NEET-UG

Botany
Module

Designed for Self-Study & Rapid Revision

SOHAN KUMAR

THE LINEAR BACKBONE



NEET Trends

- **Weightage:** **HIGH**
- **Question Types:**
 - Diagram Labeling (ETS)
 - Calculations (ATP/RQ)
 - Logic Statements (Location)

Syllabus Map:

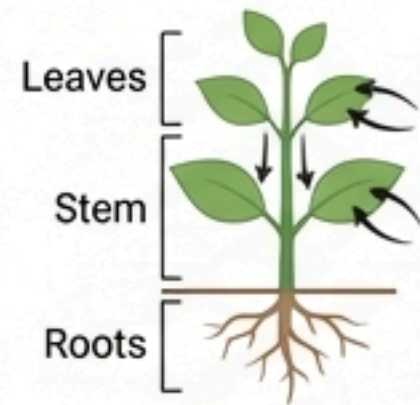
Cellular respiration – glycolysis, fermentation (anaerobic), TCA cycle, electron transport system (aerobic); energy relations; amphibolic pathways; respiratory quotient.

Do Plants Breathe? (Gas Exchange Mechanisms)

Core Question: Why do plants lack specialized respiratory organs?

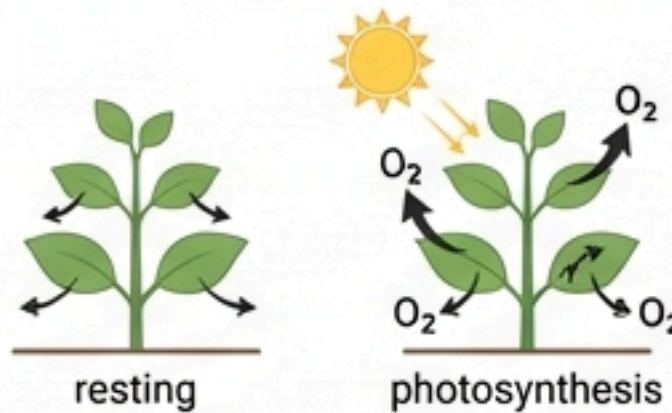
1. Independence

Each plant part takes care of its own gas-exchange needs. There is very little transport of gases from one plant part to another.



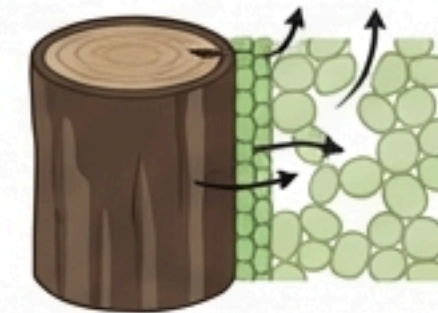
2. Low Demand

Roots, stems, and leaves respire at slow rates. High demand only occurs during photosynthesis, where O2 is produced internally.

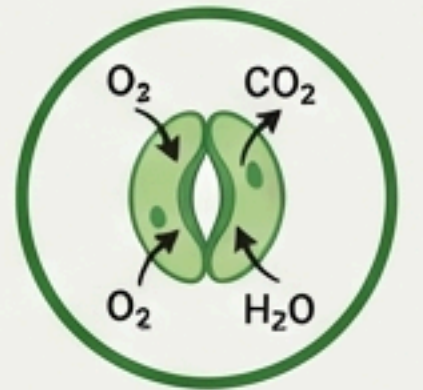


3. Superficial Location

Living cells are located close to the surface. In stems, living cells are thin layers beneath the bark. Parenchyma is loosely packed.



Stomata
(Leaves)

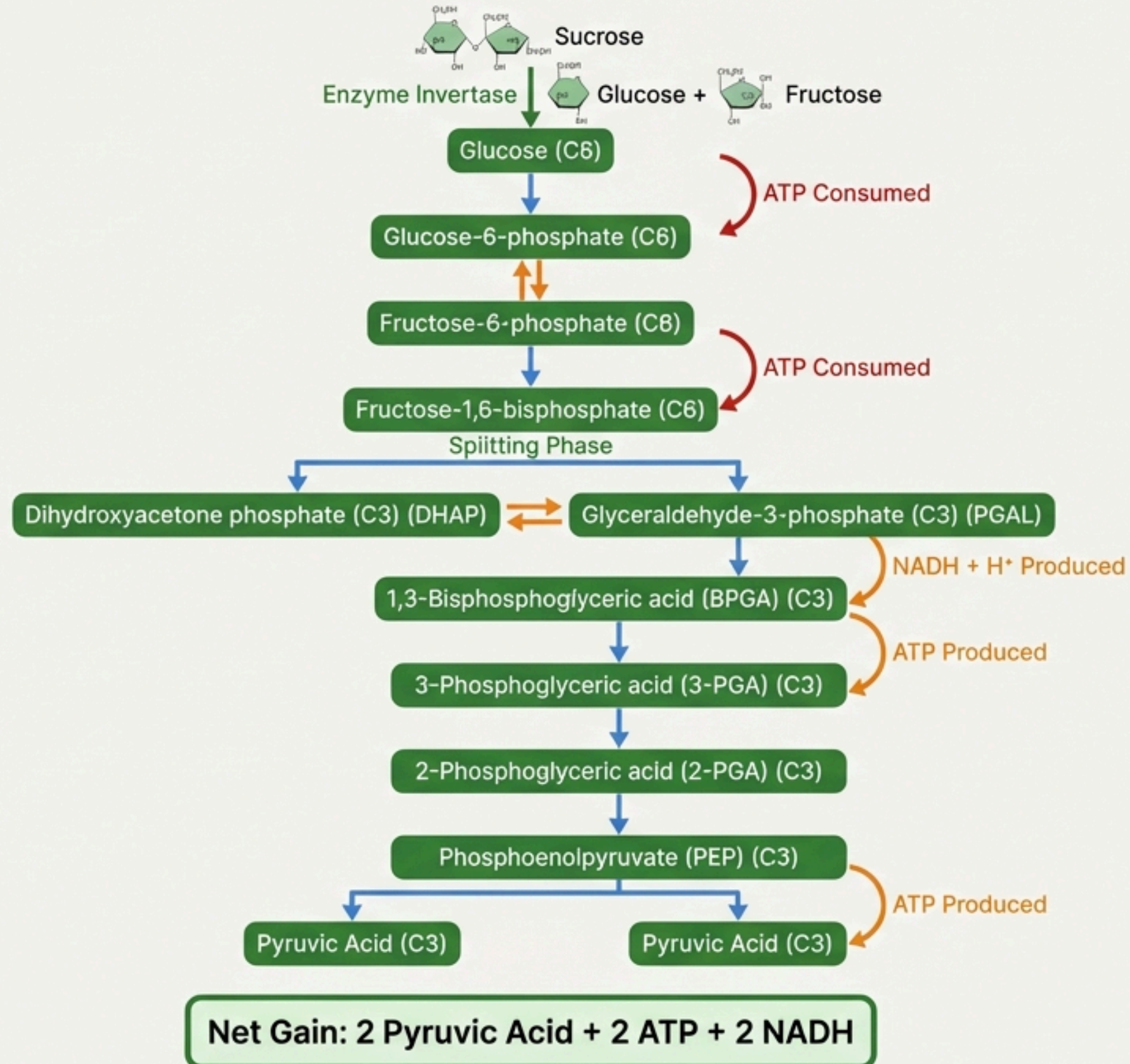


Lenticels
(Woody Stems)



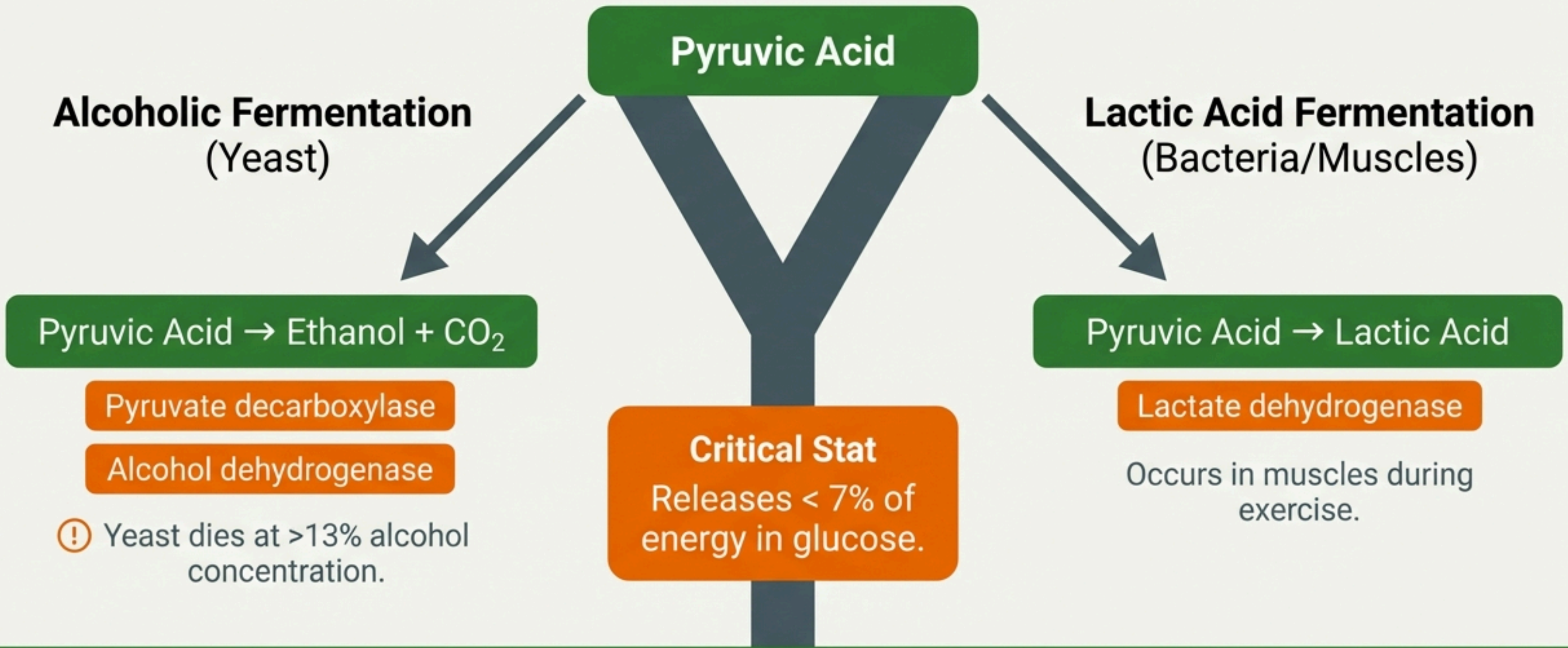
Glycolysis (EMP Pathway)

Location: Cytoplasm | Universal to all living organisms



Fermentation (Anaerobic Respiration)

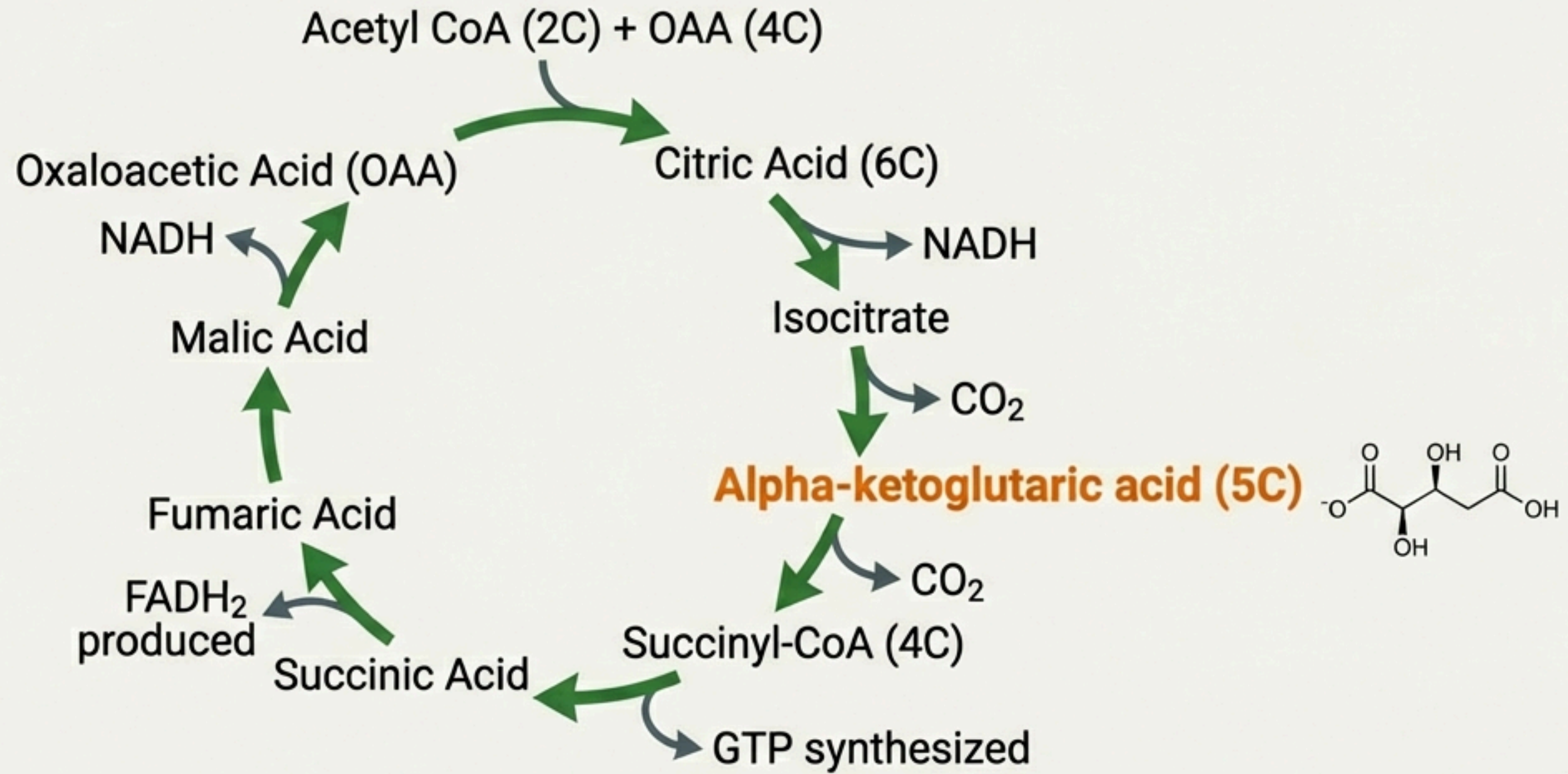
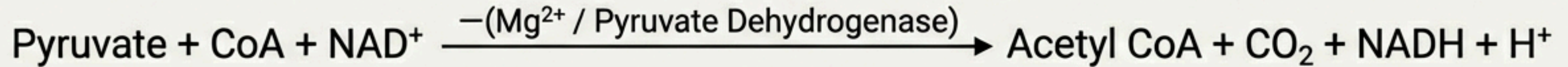
Incomplete oxidation of glucose under anaerobic conditions.



Reducing agent (NADH + H⁺) is reoxidized to NAD⁺ in both processes.

Aerobic Respiration: Link Reaction & TCA Cycle

The Link Reaction (Matrix)



Yield per Glucose (2 turns): 6 NADH, 2 FADH₂, 2 GTP, 4 CO₂

Electron Transport System (ETS)

Location: Inner Mitochondrial Membrane

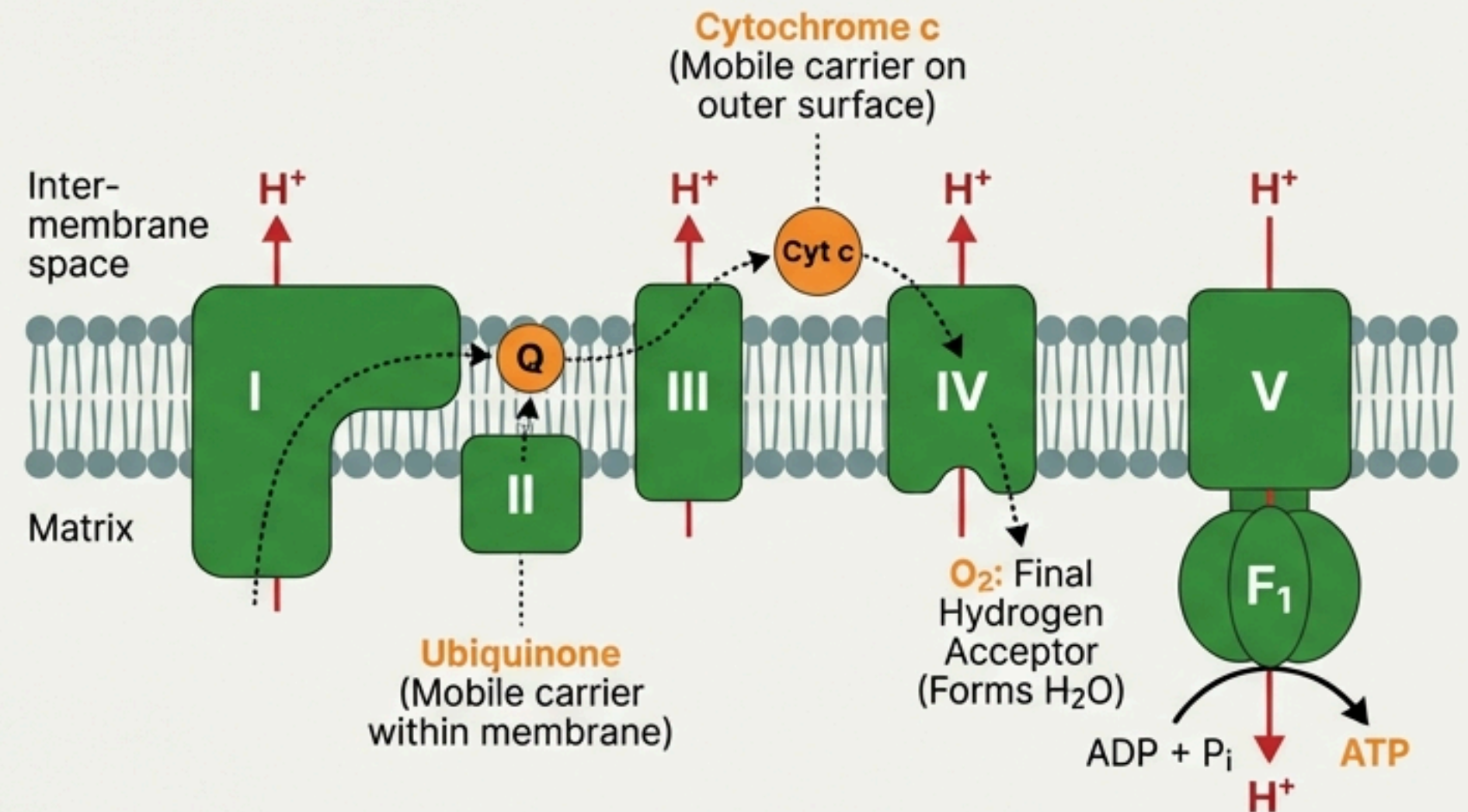
Complex I: NADH Dehydrogenase
(Oxidizes NADH)

Complex II: Succinate Dehydrogenase (Oxidizes FADH₂)

Complex III: Cytochrome bc₁

Complex IV: Cytochrome c Oxidase
(Contains Cu centers)

Complex V: ATP Synthase



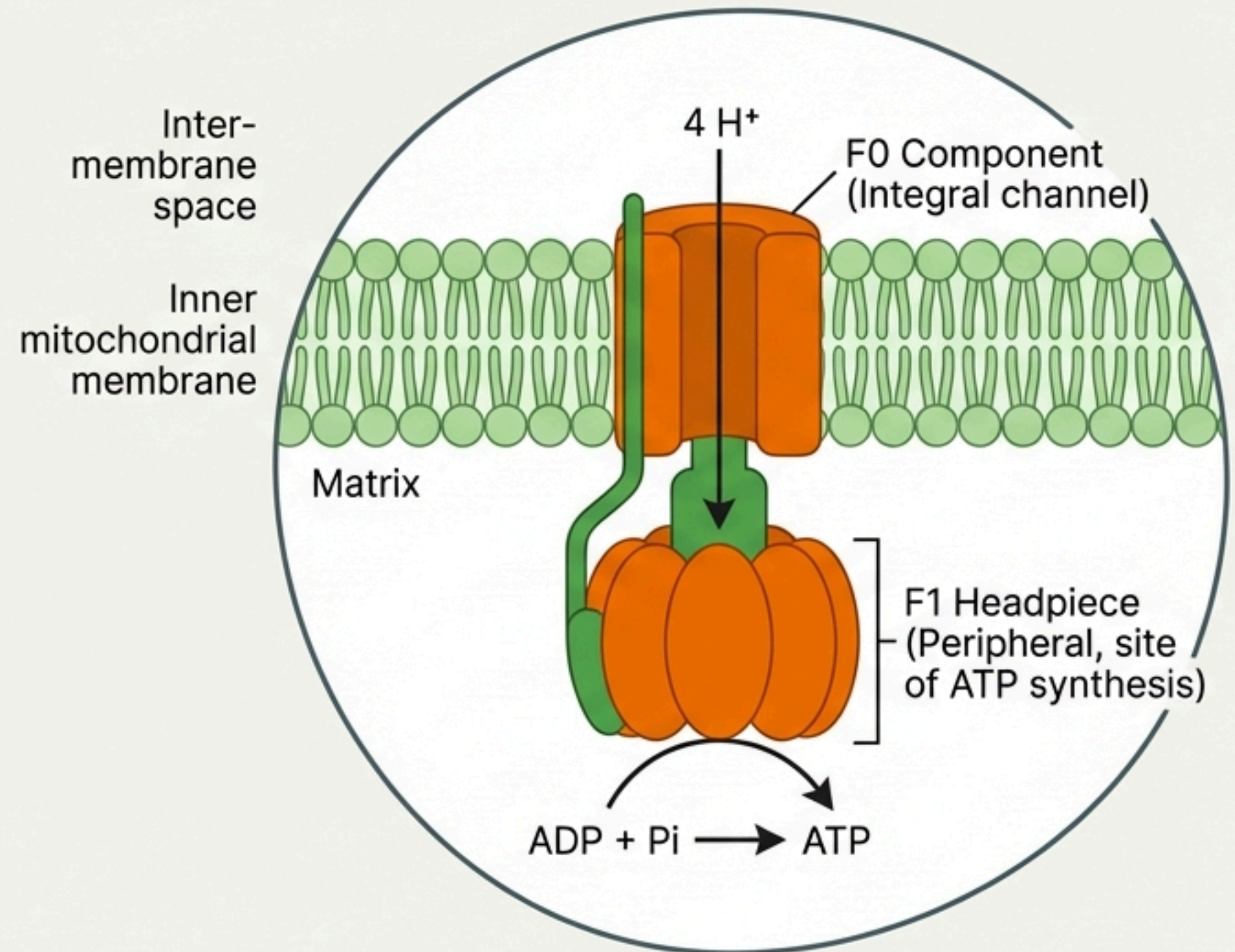
Oxidative Phosphorylation & ATP Synthase

Chemiosmotic Hypothesis

Energy of oxidation-reduction creates a Proton Gradient.

Conversion Ratios:

- 1 NADH $\rightarrow$ 3 ATP
- 1 FADH₂ $\rightarrow$ 2 ATP



The Respiratory Balance Sheet

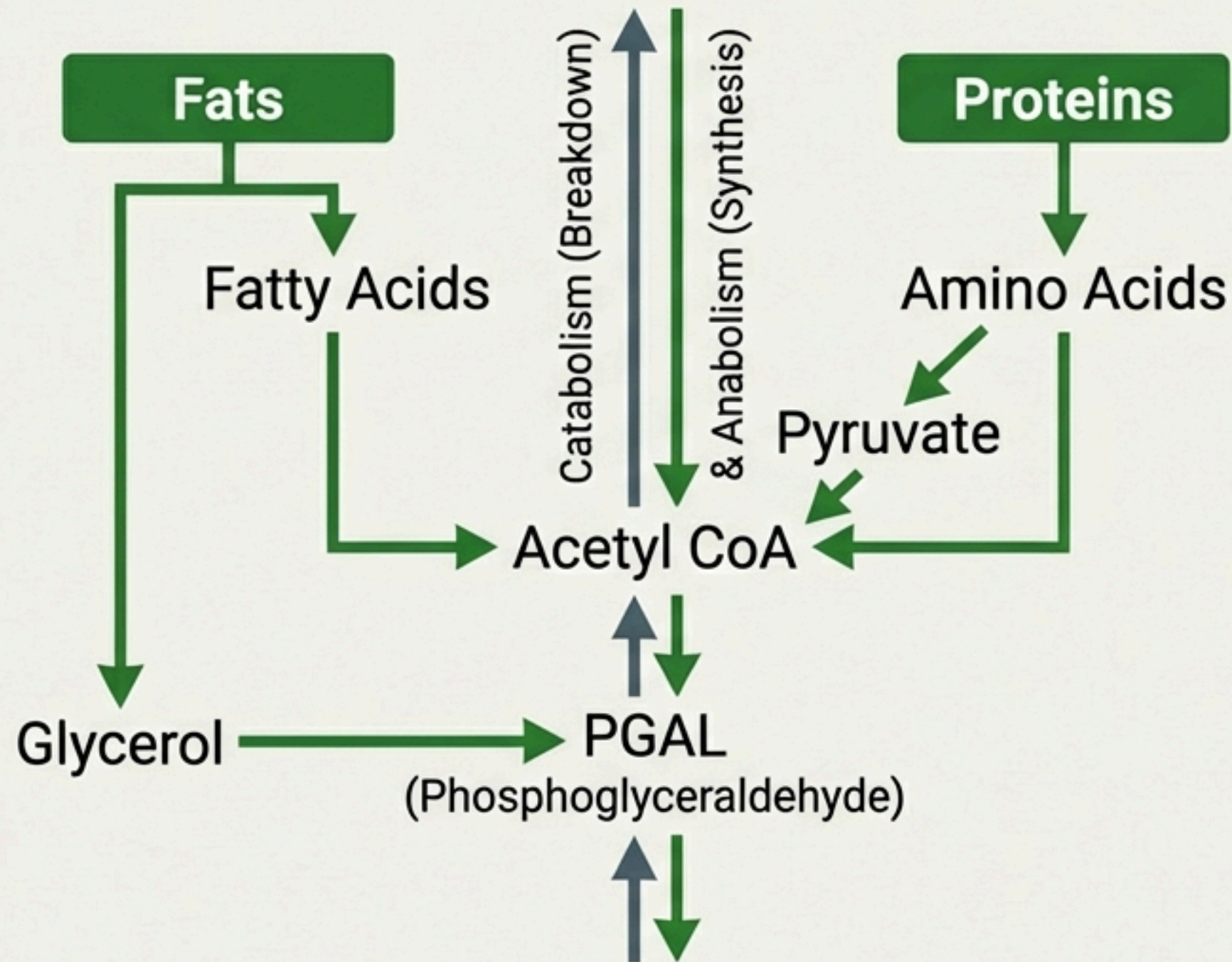
Theoretical Net Gain: 38 ATP

The Balance Sheet			
Source	Direct ATP	Reduced Coenzymes (ATP Value)	Total ATP
Glycolysis	2 ATP	2 NADH (6)	8 ATP
Link Reaction	0	2 NADH (6)	6 ATP
Krebs Cycle	2 GTP	6 NADH (18) + 2 FADH ₂ (4)	24 ATP
GRAND TOTAL			38 ATP

NCERT Assumptions

1. Sequential pathway.
2. NADH transferred to mitochondria.
3. No intermediates withdrawn.
4. Only glucose respired.

Amphibolic Pathway & Respiratory Quotient (RQ)

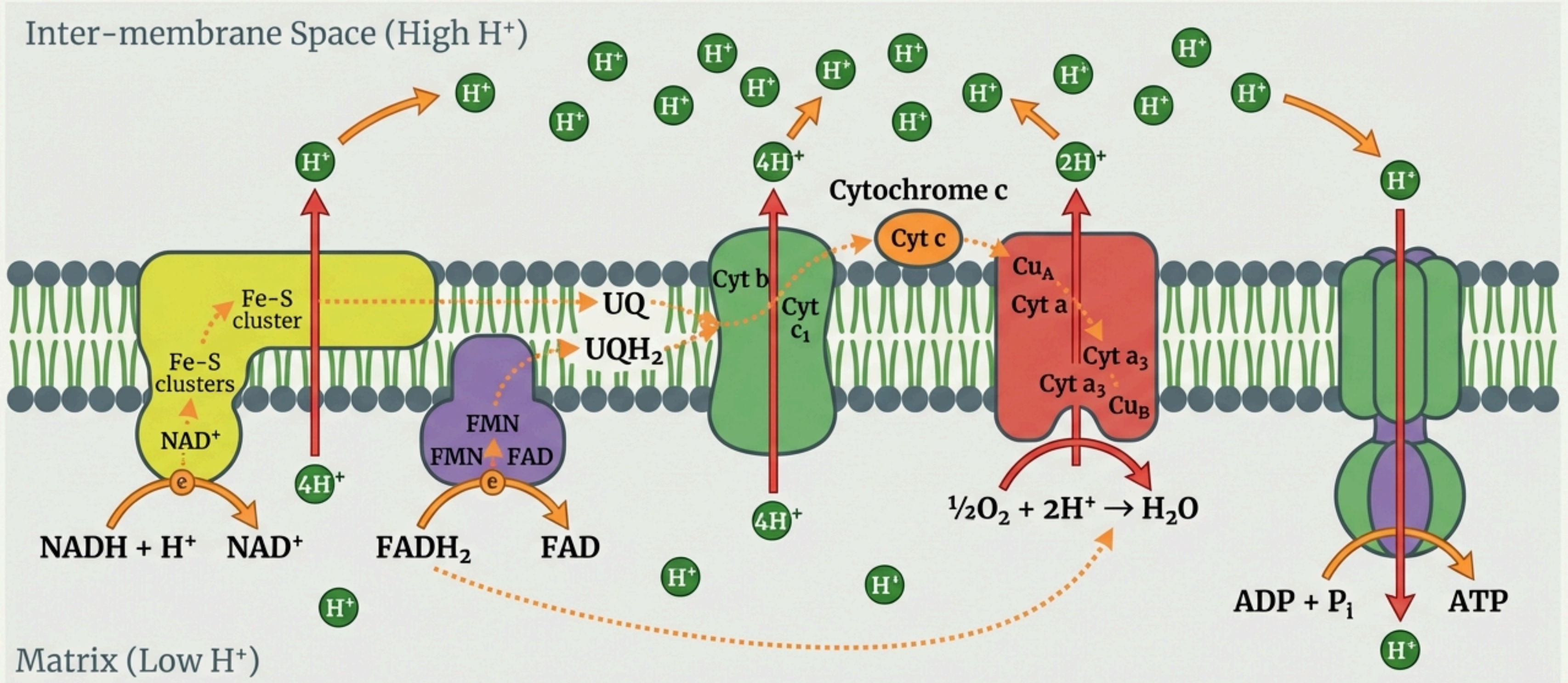


Respiratory Quotient (RQ) Values

$$RQ = \frac{\text{Vol CO}_2 \text{ evolved}}{\text{Vol O}_2 \text{ consumed}}$$

- Carbohydrates: **RQ = 1.0**
- Fats (Tripalmitin): **RQ = 0.7**
- Proteins: **RQ = 0.9**
- Organic Acids: **RQ > 1**

Visual Focus: Electron Transport System



⚠ NCERT Line Emphasis (Direct Exam Pickups)

The **first cells** on **this planet lived** in an **atmosphere** that **lacked oxygen**.

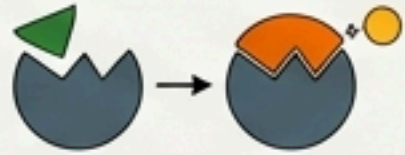
Cytochrome c is a small protein attached to the outer surface of the inner membrane.

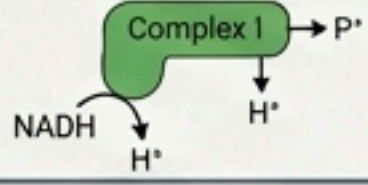
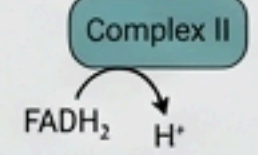
Fats would need to be **broken down** into **glycerol** and **fatty acids** first... But when the organism needs to synthesize fatty acids, **acetyl CoA** would be **withdrawn**.

Yeasts **poison themselves to death** when the concentration of alcohol reaches about 13 per cent.

Critical Comparisons

Glycolysis vs Fermentation vs Aerobic		
Process	Oxidation	Net ATP
Glycolysis	Partial	2 ATP
Fermentation	Partial	2 ATP
Aerobic Respiration	Complete	38 ATP

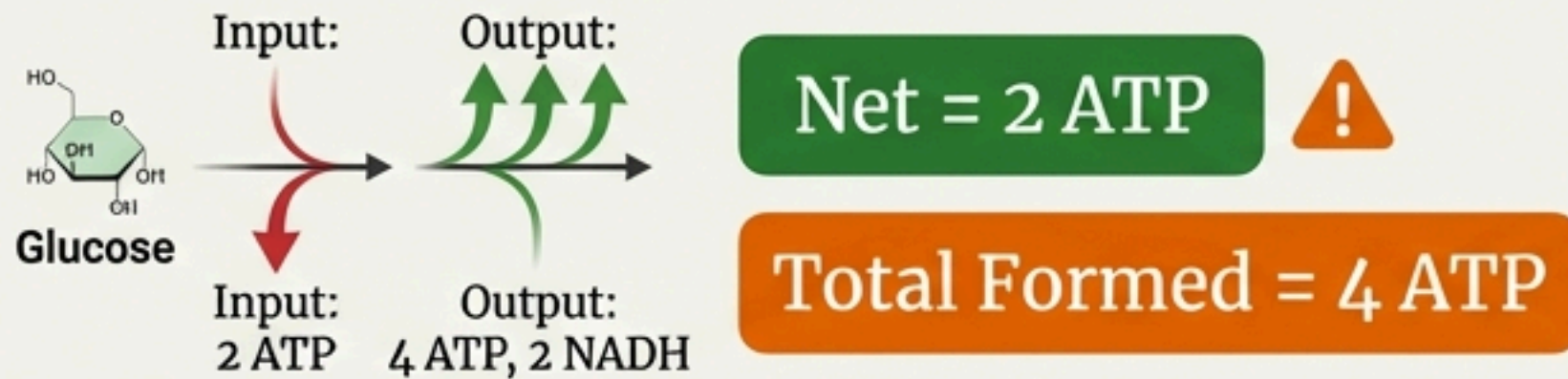
Substrate Level vs Oxidative Phosphorylation		
Substrate Level Phosphorylation	Direct transfer of phosphate group to ADP. Location: Cytoplasm/Matrix.	
Oxidative Phosphorylation	Energy from electron gradient (ETS) via ATP synthase. Location: Inner Membrane.	

Complex I vs Complex II		
Complex I	NADH Dehydrogenase (NADH input)	
Complex II	Succinate Dehydrogenase (FADH2 input)	

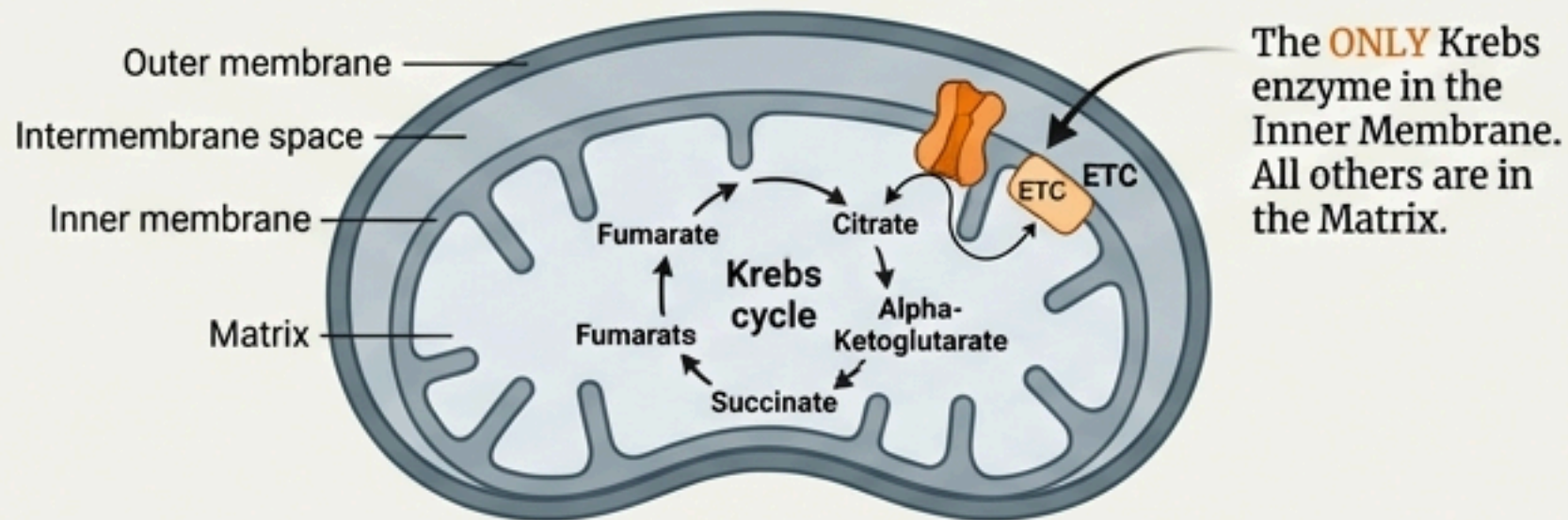
NEET-UG Focus: Traps & Concept Checks

Common Traps

Net Gain vs Total Production in Glycolysis



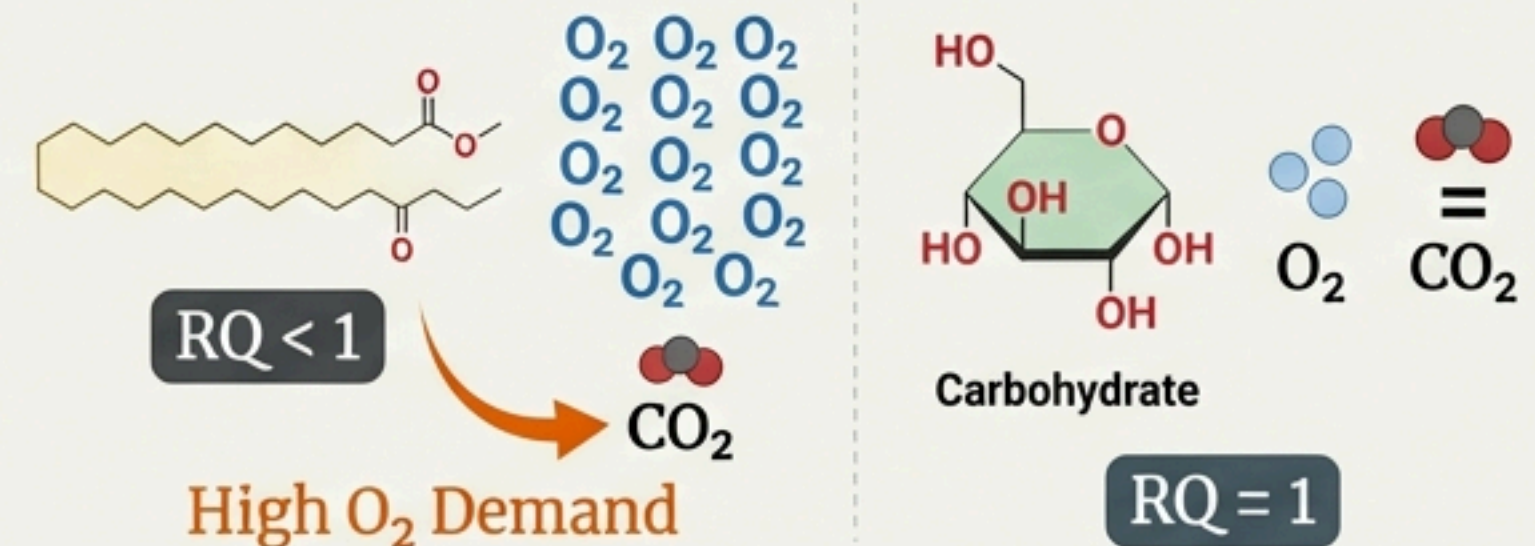
Location of Succinate Dehydrogenase



Assertion-Reason Logic

A: The RQ of fats is less than 1.

R: Fats are oxygen-poor substrates and require more O_2 for oxidation.



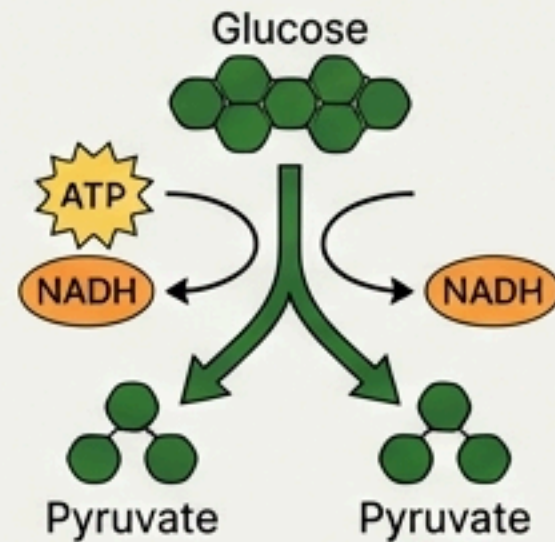
✓ **Verdict: Both True, R explains A.**

Remember: Alpha-Ketoglutaric Acid is the only 5-Carbon compound in Krebs. ⚠️

Final Chapter Snapshot

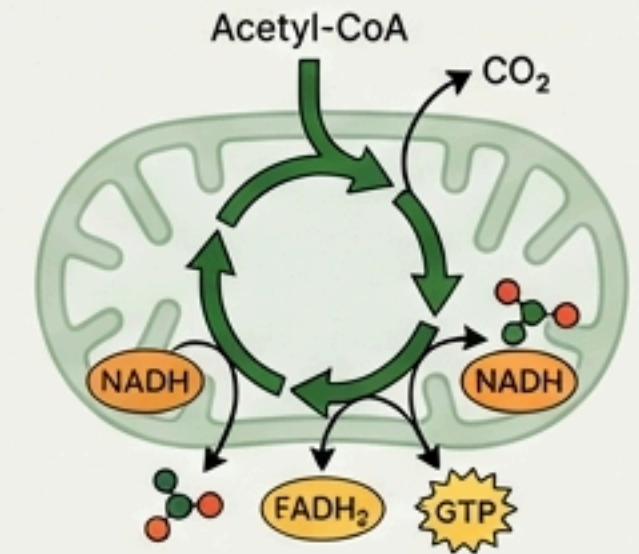
Glycolysis

- Cytoplasm
- EMP Pathway
- Net: 2 ATP + 2 NADH



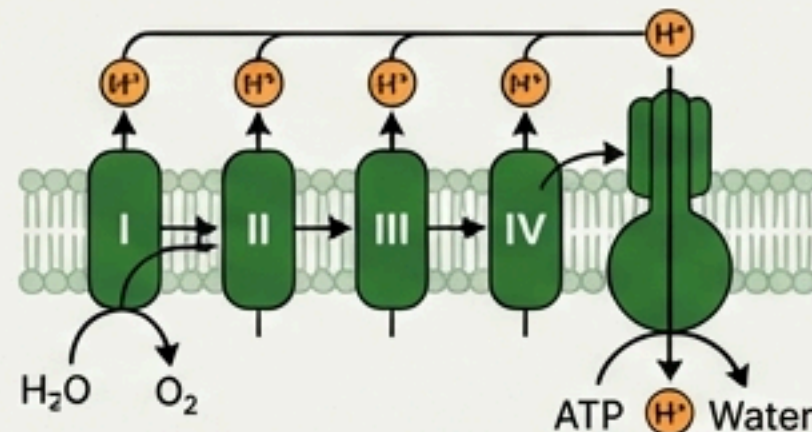
Krebs Cycle

- Mitochondrial Matrix
- Link: Oxidative Decarboxylation
- Yield: 6 NADH + 2 FADH₂ + 2 GTP



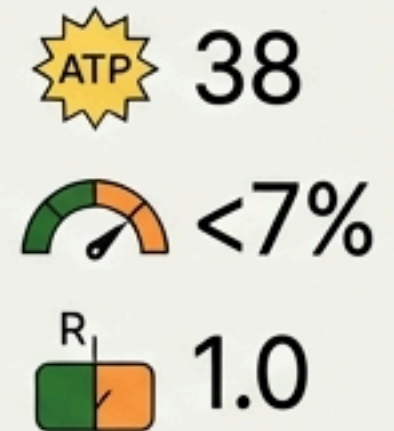
ETS

- Inner Membrane
- Complexes I-V
- Final Acceptor: Oxygen



Key Numbers

- Total ATP: 38
- Anaerobic Efficiency: <7%
- RQ Carbs: 1.0



“Respiration is an Amphibolic pathway: The central hub of carbon skeletons and energy currency.”